Algebra 1 Name____________________
Unit 5 Date____________________
Lesson 1 Homework Period____________________

1. When solving an equation, the goal is to ____________________ the variable.

Two students, James and Joe, were solving the equation $6 - 5(11 - 2x) = 17x$. Their work is shown below for questions 2 – 5.

James' Work	Joe's Work
$6 - 5(11 - 2x) = 17x$ $\frac{6 - 5(11 - 2x)}{6} = \frac{17x}{6}$ $-5(11 - 2x) = \frac{17x}{6}$ $-55 + 10x = \frac{17x}{6}$ $\underline{\quad -10x \quad -10x}$ $-55 = -\frac{43x}{6}$ $\underline{\times 6 \quad \times 6}$ $-330 = -43x$ $\frac{330}{43} = x$	$6 - 5(11 - 2x) = 17x$ $6 - 55 + 10x = 17x$ $-49 + 10x = 17x$ $\underline{\quad -10x \quad -10x}$ $\frac{-49}{7} = \frac{7x}{7}$ $-7 = x$

2. Describe the different first steps the students used in solving the equation.

3. Which student made an error?

4. Write a note to the student who made an error explaining the mistake they made.

5. Is 8 a solution to the equation $3(x-2)+5=23$? Justify your answer mathematically.

For questions 6 – 10, solve each equation and verify the solution using substitution.

6. $-5(2r-0.3)+0.5(4r+3)=-64$

7. $\frac{3m+5}{3}=9m+2$

8. $\frac{2}{3}+13k-1=3k-\frac{5}{3}$

9. $4.1-7.2z+3z=5.2+z$

10. $\frac{1}{4}x - 3(x - 2) = -2.25x + 1$